

Project Meeting Notes

Date: 10 January 2026

Time: 10:00 AM – 11:30 AM

Meeting Type: Weekly Project Review

Mode: Online (Google Meet)

Attendees

- Project Manager
- Backend Developer
- Data Engineer
- Machine Learning Engineer
- QA Engineer

Meeting Objective

The objective of this meeting was to review the current progress of the project, discuss technical and operational challenges, align on upcoming milestones, and document decisions related to system performance and data handling.

Context

This project focuses on building a speech-to-text system capable of processing audio data from multiple sources, including uploaded files and online media. The team is currently transitioning from prototype development to system optimization and validation.

Detailed Discussion Notes

1. Project Progress Overview

The project manager provided an overview of the progress made during the last sprint. The data ingestion pipeline is now stable and capable of handling multiple audio formats. Initial integration with the speech-to-text engine has been completed, and transcripts are being generated successfully for clean audio inputs.

2. Technical Challenges

The backend team highlighted latency issues during real-time transcription, especially for longer audio streams. Memory usage spikes were also observed when handling concurrent requests. The ML engineer mentioned that background noise and overlapping speech significantly reduce transcription accuracy.

3. Data & Model Discussion

The data engineer discussed the importance of validating audio quality before processing. Suggestions were made to introduce preprocessing steps such as noise reduction and silence trimming. The ML engineer proposed fine-tuning the model using domain-specific speech samples to improve accuracy.

4. Testing & Quality Assurance

The QA engineer shared early testing results and noted inconsistencies in transcripts generated from accented speech. Edge cases such as very short audio clips and incomplete sentences were identified for further testing.

Decisions, Action Items & Next Steps

Key Decisions

- Implement audio preprocessing as a mandatory step before transcription.
- Prioritize optimization of real-time transcription latency.
- Use publicly available datasets with verified transcripts for benchmarking accuracy.

Action Items

1. Optimize backend API performance – Backend Developer
2. Add noise reduction and silence trimming – Data Engineer
3. Fine-tune the speech model with clean datasets – ML Engineer
4. Expand test coverage for edge cases – QA Engineer

Risks & Mitigation

Potential risks include degraded accuracy for noisy audio and scalability issues under high load. These risks will be mitigated through improved preprocessing, model fine-tuning, and load testing.

Next Meeting

The next review meeting is scheduled for 17 January 2026 at 10:00 AM. The focus will be on performance metrics, transcription accuracy, and deployment readiness.